1 A company designs and builds tablet computers. The company wants to reduce the cost of manufacturing their tablet computers. The company also decides to stop releasing software updates for their older tablet computers. This means customers will need to change their devices more often.

Discuss the impact of these decisions on the company and the customers.

Include in your answer:

- ethical issues
- environmental issues
- benefits and drawbacks.

© OCR 2026. You may photocopy this page.

Created in ExamBuilder

[8]

2 A youth centre is creating a network with 20 computers in a single building that will be connected in a Local Area Network (LAN).

Identify **one** piece of legislation that the youth centre will need to follow when storing users' data **and** describe the steps they need to take to meet the requirements of this legislation.

Legislation

Steps the youth centre needs to follow

[4]

© OCR 2026. You may photocopy this page.

Created in ExamBuilder

3 A computer programmer has developed a computer game that they want to release for users to download over the internet. The programmer needs to decide whether to release the game as open source or proprietary software.

Discuss the features, benefits and drawbacks of each type of licence for this program and make a recommendation to the programmer.

You should include the following in your answer:

- features of each licence
- legal and ethical issues of each licence
- benefits and drawbacks of each licence.

[8]

4 An artist is working with a programmer on the development of a new piece of software.

The software will allow users to edit images on devices such as mobile telephones.

They are considering releasing the software as open source instead of proprietary.

i. Describe **two** benefits to the artist and programmer of releasing the software as proprietary.

1 _____

_____

_____

_____

2 _____

_____

_____

_____ **[4]**

ii. Describe **one** benefit to the users of releasing the software as open source.

_____

_____

_____

_____ **[2]**

© OCR 2026. You may photocopy this page.

Created in ExamBuilder

5 A shopping centre has a security system that includes CCTV cameras to record activities in the centre. The security system is being upgraded to include the use of facial recognition to identify, track the movements of and record individuals throughout the shopping centre.

Discuss the positive and negative impacts of this upgrade including:

- ethical issues
- privacy issues
- legal issues

[8]

6 *Social networking websites use artificial intelligence (AI) to monitor posts from users.

Discuss the positive and negative uses of AI by social networking websites including:

- Legal issues
- Ethical issues
- Privacy issues

[8]

7 Layla uses her computer to create educational games.

When Layla has finished her educational game, she is going to release it as open source.

Give **one** benefit and **one** drawback of Layla releasing her game as open source.

Benefit

Drawback

[2]

END OF QUESTION PAPER

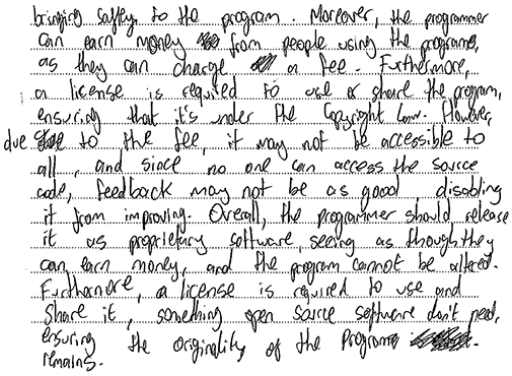
Question			Answer/Indicative content	Marks	Guidance
1			**Mark Band 3–High Level** **(6-8 marks)** The candidate demonstrates a thorough knowledge and understanding of a wide range of considerations in relation to the question; the material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate is able to weigh up both sides of the discussion and includes reference to the impact on all areas showing thorough recognition of influencing factors. *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated*. Includes benefits and drawbacks. At least one ethical and at least one environmental issue. References both customer and company clearly for top of band. Points are expanded. **Mark Band 2-Mid Level** **(3-5 marks)** The candidate demonstrates reasonable knowledge and understanding of a range of considerations in relation to the question; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate makes a reasonable attempt to discuss the impact on most areas, showing reasonable recognition of influencing factors. *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence*. Covers benefits or drawbacks to achieve band, top of band requires at least one of each. Covers ethical and/or environmental. Some points are expanded. May only reference customer or company.	8	The following is indicative of possible factors/evidence that candidates may refer to but is not prescriptive or exhaustive: **Indicative Content:** Ethical issues: • Will cost remain the same? – unfair on customers • Will customers be told? Or are they being deceived? • Can improve profit for company – allow staff to be paid more • Ethical to the company but unethical to the customers • Reduced cost may mean reduced reputation- long term loss If people move to other companies • Customers want most up-to-date devices – prepared to spend regularly • Lower cost to purchase may allow more people to afford device Environmental issues • Increase in e-waste from products that are disposed • Older devices could be distributed to people without devices to reduce waste • Products could be recycled for components Benefits and drawbacks: • Company Increase sales Increase profit Decrease popularity/reputation • Customer Increase in cost May be able to buy when couldn't before if initial cost is less Less reliable devices Effort/time to install new devices more regularly <u>Examiner's Comments</u> Many responses were suitably structured, with the stronger responses taking each of the bullet points in the question in turn. Some responses made appropriate use of bullet points and tabular structures, especially for the benefits and drawbacks which presented their arguments appropriately. When using bullets and tables candidates need to make sure they include their expansions alongside.

Question			Answer/Indicative content	Marks	Guidance
			Mark Band 1-Low Level **(1-2 marks)** The candidate demonstrates a basic knowledge of considerations with limited understanding shown; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion. *The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.* Has a point for ethical/environmental/benefit/drawback, not all relevant. May not reference customer or company. **0 marks** • No attempt to answer the question or response is not worthy of credit		Candidates often focused on the drawbacks of the decision, most commonly the environmental impact of the waste generated from replacing devices. Similarly, many responses identified a potential increase in the digital divide. The most common argument was that people could not always afford to purchase new devices, and so would not have access to the most up-to-date technology.. Fewer responses considered a range of potential benefits beyond the increased profit for the company. Some stronger responses identified recycling the old devices to avoid waste, and providing these older devices at a lower cost to people who could not afford the newest devices. Exemplar 1 ~~By releasing~~ By not releasing software updates for their older tablet computers, meaning customers will need to change devices more often a negative impact on the environment happens because more waste is produced such as batteries and plastics that go into landfill and pollutes the planet. The company will generate greater profit from reducing the manufacturing cost and also forcing customers to purchase new devices more frequently. However the company may lose business if the user decides that they dislike the ethics of the company polluting the planet. Business may also decrease if the customer isn't satisfied that their tablet computer is still good. The customer may get unhappy that a new model has been released so soon. Reducing manufacturing costs is good as long as it's done ethically and doesn't take advantage of any workers or people in labour. Having a planned lifespan for a device is called planned obsolescence. Overall, the benefits are a greater profit and the company can dedicate more resources to developing new technology. However the drawbacks are environmental damage, a worsening company reputation is possible, the risk of losing business and the ethical implications of ensuring that manufacturing is carried out ethically and not taking advantage of less fortunate people. In my opinion I would advise against this change as the risks are very large and outweigh the potential positives for the company. This candidate considers both the benefits and drawbacks. They identify the environmental drawbacks of extra pollution and that customers may not be happy at having to replace devices so often. The response also considers the benefits of the company increasing their profits.

Question			Answer/Indicative content	Marks	Guidance
					This is expanded to explain how the profit can lead to more benefits, that they could put more resources into developing new technology. This response gained the top band mark because they included ethical and environmental impacts. They gave benefits and drawbacks. The response had expansions; the candidate did more than stating a point, they explained why it was relevant, or how this was a drawback. They also considered both the customer and the company and how the decision would affect both.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
2			1 mark for legislation Data Protection Act (2018) // DPA // GDPR // Data protection 1 mark each for steps to max 3 e.g. • Identify data protection officer • Keep data secure/safe (from unauthorised access) • … by example // installing firewall // usernames and passwords // access rights // physical security // encrypt • Do not share/sell (without consent) • Don't store without consent • Only keep relevant data • … do not collect data from users that is not required • Do not store the data for longer than needed • Make sure the data remains accurate / up-to-date • Only use for purpose collected // Data must not be misused • … state purpose for collection • Do not store/share with countries with lower levels of protection • Change/delete data when requested … • … permanently/securely • Provide data for users when they request to see it • Notify individuals of data breaches • Must use data lawfully/fairly • Staff need to be trained in principles	4	If no legislation, wrong legislation, unclear legislation still award up to max 3 marks for description of DPA. Accept answers listing what needs to be done e.g. data must be up-to-date (without the specific step that will be followed to do this)? MP2 is stopping unauthorised access. MP4 is the company actively giving the data willingly to someone else. <u>Examiner's Comments</u> Responses often correctly identified the data protection act or GDPR as being the legislation. Some identifications were descriptions of legislation, for example to keep data secure, instead of the actual name of the legislation. Candidates commonly identified the need to keep the data secure and often gave a suitable method for this such as encrypting the data. Some responses also gave detailed steps that the youth centre would need to follow, such as allowing access when requested, deleting data when no longer needed, making sure that the data is kept up to date and not sharing the data without consent. Some candidates inaccurately identified the Computer Misuse Act or Copyright Designs and Patents Act, which are not followed when storing user's data.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
3			**Mark Band 3–High Level** **(6-8 marks)** The candidate demonstrates a thorough knowledge and understanding of a wide range of considerations in relation to the question; the material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate is able to weigh up both sides of the discussion and includes reference to the impact on all areas showing thorough recognition of influencing factors. *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* The answer covers all required elements (legal/ethical, benefits, drawbacks) given in the question about open source **and** proprietary and includes a recommendation with justification. The top of the band makes a clear and structured recommendation to the programmer. **Mark Band 2-Mid Level** **(3-5 marks)** The candidate demonstrates reasonable knowledge and understanding of a range of considerations in relation to the question; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. **The candidate makes a reasonable attempt to discuss the impact on most areas, showing reasonable recognition of influencing factors.** *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* The answer includes one or more from legal/ethical, benefits, drawbacks for open source and proprietary. Alternatively, the answer could have a justified recommendation without clearly	8 AO2 1a (4) AO2 1b (4)	The following is indicative of possible factors/evidence that candidates may refer to but is not prescriptive or exhaustive: **Indicative Content:** Licence features Open source – (usually free), can access/change source code, redistribute Proprietary – purchase at a cost, cannot access/change code Legal and ethical issues: • Both provide copyright • Open source – allows more people to take code and possibly change to resell, or adapt in their own programs to resell or claim as their own (reverse for proprietary) • Open source – allows more people access to the game because there is likely no cost (reverse for proprietary) Benefits and drawbacks: • Open source – wider customer base, more exposure, users can alter to make it better/fix bugs, limited documentation, little financial gain • Proprietary – allows programmer to earn money, gives more control over what happens with the program, usually well tested, more restrictions for copyright, cannot be adapted to meet user needs, Decision: Either would be appropriate, justification needs to be clearly for the scenario <u>Examiner's Comments</u> This question required candidates to provide an extended response. An extended response can be given in the form or paragraphs, key points as well as a table of points. Candidates need to make sure they are covering the three bullet points in the question for both the open source licence and proprietary. The question also asked for a recommendation to the programmer.

Question			Answer/Indicative content	Marks	Guidance
			referencing the bullet points in the question. **Mark Band 1-Low Level** **(1-2 marks)** The candidate demonstrates a basic knowledge of considerations with limited understanding shown; the material is basic and contains some inaccuracies. Thecandidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion. *The information is basic and communicated in an unstructured way.* *The information is supported by limited evidence and the relationship to the evidence may not be clear.* The answer is limited to the facts about open source **and/or** proprietary. **0 mark** No attempt to answer the question or response is not worthy of credit		Candidates were often able to identify the features of each licence, for example if the source code was provided. Candidates often included benefits and drawbacks, for example being able to edit the program to tailor it to their needs, the potential of misuse of the program code. Candidates often covered legal and ethical issues within their benefits and drawbacks without explicitly identifying them. Fewer candidates included a recommendation for the programmer. Candidates described each in turn without identifying which one should be used. Some candidates suggested that both were suitable and it was the programmer's decision, but the question asked for a recommendation. The stronger responses discussed each licence in turn and then in the final paragraph started with a clear recommendation and justified the reasons for this by providing a summary of the points they had discussed in detail previously. **Exemplar 1**  bringing safety to the program. Moreover, the programmer can earn money from people using the programme, as they can charge a fee. Furthermore, a license is required to use & share the program, ensuring that it's under the copyright law. However, due to the fee, it may not be accessible to all, and since no one can access the source code, feedback may not be as good, disabling it from improving. Overall, the programmer should release it as proprietary software, seeing as through they can earn money, and the program cannot be altered. Furthermore, a license is required to use and share it, something open source software don't need, ensuring the originality of the programme remains. This response has a clear recommendation at the end of the response. They have stated that the programmer should use proprietary and provided a summary of the reasons (discussed previously) as to why they think this is the most appropriate recommendation.
			Total	**8**	

Question			Answer/Indicative content	Marks	Guidance
4		i	1 mark each: e.g. • Authors can earn money • …by selling for a fee // using licences to stop unauthorised use • No-one can see the code • …users cannot edit/change the software // by example inserting malware • …so it cannot be copied/resold/shared • More **control** over intellectual property // by example e.g. restrict users, restrict what can be done with the software without permission	4	Benefit is to artist and programmer – not user. Mark the answer as a whole. Do not award reference to ownership/copyright because both allow copyright of the code/program. <u>Examiner's Comments</u> This question required candidates to consider the benefits to the developer of releasing the software as proprietary. A common error was describing the benefits to the users, for example they know the software has not been tampered with and that there is lots of support. The most common responses included the ability to earn money from selling the software and retaining control due to no-one else being able to access the source code.
		ii	1 mark for point, 1 for expansion: e.g. • Users can **view/edit** the (source) code // Users can **edit** the program/software • …to **tailor/improve/adapt** it to do what they need/want • …so errors can be fixed (by anyone) • …users can learn how the software works • Freely accessible • …do not have to pay • …can redistribute … • …with changes	2	Benefit to users not artist and programmer <u>Examiner's Comments</u> This question required candidates to consider the benefits to the users of the software being released as open source. Many candidates were able to accurately identify that the source code comes with the software, and this allows the user to edit it to meet their needs.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
5			**Mark Band 3–High Level** **(6–8 marks)** The candidate demonstrates a thorough knowledge and understanding of a wide range of considerations in relation to the question; the material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate is able to weigh up both sides of the discussion and includes reference to the impact on all areas showing thorough recognition of influencing factors. *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Mark Band 2-Mid Level** **(3–5 marks)** The candidate demonstrates reasonable knowledge and understanding of a range of considerations in relation to the question; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate makes a reasonable attempt to discuss the impact and most are showing reasonable recognition of influencing factors. *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Mark Band 1-Low Level** **(1–2 marks)** The candidate demonstrates a basic knowledge of considerations with limited understanding shown; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion.	8 AO2 1a (4) AO2 1b (4)	The following is indicative of possible factors/evidence that candidates may refer to but is not prescriptive or exhaustive: **Indicative Content:** Some points may cover more than one 'issue'. Legal issues: • DPA needs to be followed or company could be fined e.g. Customers informed the system is used Data held for specified time/reasons Data kept secure • Centre is private property so customers can choose not to enter • Can be used to identify people committing crimes e.g. theft, used as evidence, make sure the correct people are caught. Ethical issues: • Users feel safer because they know any actions are being monitored and help/action will be taken if needed • If users have not done anything then there is no reason to be tracked/recorded so should not impact them • Users feels unsafe because they are being watched • Users may be unaware they are being recorded - need to be informed, give consent • Users do not know where the videos/data about them and their movements is stored/how it is used - DPA reference Privacy issues: • Users may feel it is an invasion of privacy • Users are in a public place and can be legally recorded by anyone anyway • Users may feel like they are being watched all the time • Users have not given their permission to be tracked • Users may not know the system exists <u>Examiner's Comments</u> The quality of written communication

Question			Answer/Indicative content	Marks	Guidance
			The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 mark No attempt to answer the question or response is not worthy of credit		required a balanced discussion of the positive and negative impacts of including facial recognition with CCTV cameras in a shopping centre. Some responses gave strongly negative arguments with little, if any, consideration for the positive impacts. Some candidates also chose to focus on the introduction of CCTV cameras, where the question states there are already CCTV cameras and the discussion is about the upgrade to facial recognition. The more successful responses considered the ethical, privacy and legal issues one at a time and identified the positive and negative impacts (where applicable) for each of these three sections. These responses were usually well structured and had a balanced discussion before leading to justified conclusions. Some successful responses included bullet points and table structures to help the candidates structure their response. These were often thorough and included detailed discussions. Some of the less successful responses focused on people not wanting to be watched by CCTV cameras, or that people would misuse the CCTV cameras. The CCTV cameras were already in existence and therefore this was irrelevant to the context of the question. More successful responses considered the storage of the tracked videos and how this could lead to data privacy issues. These responses also discussed how shops could use the stored data to analyse where people go in shops to identify areas where they could put more adverts.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
6			**Mark Band 3–High Level** **(6-8 marks)** The candidate demonstrates a thorough knowledge and understanding of a wide range of considerations in relation to the question; the material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence / examples will be explicitly relevant to the explanation. The candidate is able to weigh up both sides of the discussion and includes reference to the impact on all areas showing thorough recognition of influencing factors. *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Mark Band 2-Mid Level** **(3-5 marks)** The candidate demonstrates reasonable knowledge and understanding of a range of considerations in relation to the question; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate makes a reasonable attempt to discuss the impact on most areas, showing reasonable recognition of influencing factors. *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Mark Band 1-Low Level** **(1-2 marks)** The candidate demonstrates a basic knowledge of considerations with limited understanding shown; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion. *The information is basic and communicated in an unstructured way.*	8 (AO2 1a (4)) (AO2 1b) (4))	The following is indicative of possible factors/evidence that candidates may refer to but is not prescriptive or exhaustive: **Indicative Content:** Legal issues: • Copyright designs and patents act - can check for plagiarism automatically and highlight posts e.g. videos or images • Data protection act - needs to make sure rules are followed so that the AI algorithm does not breach e.g. security • Check that materials are all legal • User has agreed the terms when signing up so should expect it Ethical issues: • Users may not want everything they post monitoring • May incorrectly block users/posts • Can limit plagiarism • Can make sure inappropriate/illegal posts are not published • Website will need to tell users what it is doing and they must agree with it • Record of monitoring reports may be stored and used for other means • Users may feel safer using the website because they know inappropriate material will not be published Privacy issues: • Users may feel like they are being watched all the time • Terms and conditions may sign away their rights to privacy when using the website • People may prefer a computer analysing their posts than people reading them <u>Examiner's Comments</u> This question was a quality of written communication question. This means that candidates need to consider how they structure their response to make sure it has a clear narrative, and to consider their grammar and spellings throughout. This does not mean that an essay is required,

Question			Answer/Indicative content	Marks	Guidance
			The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit		some of the stronger responses used headers and bullet points that clearly identified which issues were positive, negative, related to legal, ethical or privacy. By doing this candidates were also making sure that they knew they had covered all requirements. These bullet points were not single words, they included appropriate descriptions and explanations where required (sometimes as sub-bullet points). Some responses evolved into generic answers about AI in the world, for example developing into robots that are living organisms, instead of focusing on the use of AI for monitoring posts on a social networking website. Many responses were heavily negative, identifying a lack of privacy and trust, without identifying many positive features, such as faster identification of inappropriate content. The stronger responses gave a balanced argument of positive and negatives, often identifying a use, e.g. Looking for inappropriate content, and then giving the positive expansions and then the negatives. This ensured that they had identified at least one point for legal, ethical and privacy, and had a balanced response. Exemplar 4 Using artificial intelligence to monitor posts is much more effective than a person monitoring, due to the fact that an AI can monitor multiple posts at once. A negative to using AI is that users can bypass the AI monitoring because in most cases, they look for key words or phrases to be able to punish users. This means that users can misspell words or phrases to bypass the AI There are major privacy issues which could accompany the AI, as users do not know what sort of information that a piece of artificial intelligence can access. This could lead to users feeling unsafe, ~~therefore~~ therefore they could leave the website or complain. Secondly, Ethical issues could also be a problem because users could feel as if a website is not properly ~~monitored~~ monitored by the AI if it was faulty or

Question			Answer/Indicative content	Marks	Guidance
					not properly set up. This could lead to issues with the website where certain posts can go through without being noticed by the AI. Finally, there are legal issues with letting AI monitor the posts from users because the AI could be setup for malicious purposes, which violates the Computer Misuse act, 1990. This is a negative as the website could be taken down. This response has some valid points but they are not all accurate. For example, suggesting the AI can access other data from the users would suggest it is doing something other than it is designed to do. The candidate has identified that if it is not set up it might not work correctly, but this is the same as using people to monitor the posts. They also suggest it could be set up maliciously - which is beyond the scope of the question. The question states that the AI is only used to monitor posts. The response is primarily negative and the positive points are limited in their content and expansions.
			Total	8	
7			benefit e.g. other people can improve the program drawback e.g. cannot charge a fee // other people can use her code	2	Need to be benefit/drawback to developer
			Total	2	